

Coding & Solution

Date	1 July 2026
Team ID	XXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	OptiCrop GitHub Repository
Programming Language(s)	Python, JavaScript, HTML, CSS
Framework(s) Used	Flask, scikit-learn
Key Features Implemented	Crop Recommendation, NPK and Climate Input, Input Validation, Prediction Confidence, Responsive Web Interface
Pending / Incomplete Features	Advanced analytics and location-specific advisory features (Future Enhancement)
Setup / Run Instructions	Clone the repository, install requirements, activate the Python environment, and run the Flask application.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop application is successfully developed using Flask and a trained machine-learning crop recommendation model. The code follows a modular structure, uses meaningful naming conventions, implements error handling, and is maintained through GitHub version control. The application is fully functional and meets the defined project requirements.